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ORIGINAL ARTICLE

Leaf structural traits mediating pre-existing physical innate resistance to sorghum aphid in sorghum under uninfested conditions

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Abstract

Key message **We found four indicative traits of innate immunity. Sorghum-resistant varieties had a greater trichome, stomatal and chloroplast density, and smaller mesophyll intercellular width than susceptible varieties.**

Abstract The sorghum aphid (SA), *Melanaphis sorghi* (Theobald), can severely reduce sorghum yield. The contribution of structural traits to SA resistance has not been extensively studied. Moreover, the current screening method for resistance is inherently subjective for resistance and requires infestation in plants. Quantifying the microanatomical basis of innate SA resistance is crucial for developing reliable screening tools requiring no infestation. The goal of this study was to identify structural traits linked to physical innate SA resistance in sorghum. We conducted controlled environment and field experiments under no SA infestation conditions, with two resistant (R. LBK1 and R. Tx2783) and two susceptible (R. Tx7000 and R. Tx430) varieties. Leaf tissues collected at the fifth leaf stage in the controlled environment experiment were analyzed for the epidermal and mesophyll traits using light and transmission electron microscopy. Leaf tissues collected at physiological maturity in the field experiment were analyzed for surface traits using scanning electron microscopy. Our results showed that stomatal density, trichome density, trichome length, and chloroplast density are key leaf structural traits indicative of physical innate SA resistance. We found that resistant varieties had a greater density of trichomes (39%), stomata (31%), and chloroplast (42%), and smaller mesophyll intercellular width (− 52%) than susceptible varieties. However, the chloroplast, mitochondria, and epidermal cell ultrastructural traits were ineffective indicators of SA resistance. Our findings provide the foundation for developing an objective high-throughput method for SA resistance screening. We suggest a follow-up validation experiment to confirm our outcomes under SA infestation conditions.

Keywords Chloroplast · Defense · Host plant resistance · Phenotyping · Stomatal morphology · Trichome

Abbreviations

SA	Sorghum aphid
SEM	Scanning electron microscopy
TEM	Transmission electron microscopy

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Introduction

The sorghum aphid (SA), *Melanaphis sorghi* (Theobald), is a globally distributed insect pest of sorghum (*Sorghum bicolor* (L.) Moench). Capable of severe economic loss, SA is a persistent threat to the sorghum industry (Bowling et al. 2016) and can reduce sorghum yield by ≥ 60% (van Rensburg 1973; Van den Berg 2002; Uyi et al. 2023). An inherently subjective screening protocol is the principal method of determining SA-resistant sorghums. Using visual observation, the current protocol necessitates well-trained plant scientists and entomologists working in concert to elucidate the resistance mechanisms. The limited availability of these two coinciding factors is a definitive limitation. Determining SA resistance by objectively considering the innate traits of uninfested plants will likely provide a reliable high-throughput screening methodology to facilitate hybrid development and improve the durability of resistance. These innate traits, which are morpho-anatomical characteristics of the plant tissue, are present at the time of infestation and may be better understood by the specific mode of aphid feeding.

Several successive events outline the host-settling behavior of aphids. Notwithstanding pre-alighting behavior, the process begins at the epidermis. Trichomes, stomata, and surface architecture have been identified as mechanisms of resistance in various crops (Powel et al. 2006; Sharma et al. 2021). Trichomes, in general have consistently been indicative of antixenosis (Levin 1973; Webster et al. 1994; Peter et al. 1995; Riddick and Simmons 2014; Karabourniotis et al. 2020). Structural and glandular trichomes have been well-documented as successful feeding deterrents (Avé et al. 1987; Khan et al. 2000; Sarria et al. 2010; Wang et al. 2020). Another structure in common contact with the aphid is leaf stomata, whereas stomatal density has been correlated with aphid resistance (Luo et al. 2019). Multiple species of aphids have been found to induce abscisic acid expression and therefore, stomata closures (White 1974; Guo et al. 2016).

Certain aphid species have even exhibited a propensity for a particular leaf topology and microclimate (Ibbotson and Kennedy 1959; Powel et al. 1999, 2006). The morphology of trichomes and stomata directly influences aphid settling behavior as individuals position themselves for feeding.

The mode of aphid feeding is complex. An aphid will exploit an intercellular apoplastic pathway to the vascular stele by puncturing the anticlinal junction between two cells (Taiallingii and Hogen Esh 1993; Powel and Hardie 2000; Hewer et al. 2011). The apoplastic pathway is defined as the intercellular space outside the plasma membrane. The stylet will radially move through the mesophyll until the sieve element has been tapped. Intermittent punctures of cortical cells are expected as the aphid attempts to find the sieve element. This behavior facilitates viral infections and has been a principal subject of study (Powel et al. 2006; Sharma et al. 2021; Leybourne and Aradottir 2022). Inherent traits of sorghum mesophyll have been theorized to influence the efficacy of sorghum aphid feeding (Tetreault et al. 2019).

Sorghum aphids feeding on resistant cultivars was observed spending an average of 1.72 times longer in the probing phase according to an electrical penetration graph (Tetreault et al. 2019). These results could be symptomatic of mesophyll resistance factors. Thought to antagonize and impede stylet penetration, mesophyll resistance factors restrict access to the phloem. This has been found to be the principal mechanism of resistance in numerous plant families regardless of an aphid's host range (Leybourne and Aradottir 2022). Despite our understanding of the general location of these resistance factors, the key traits that induce aphid feeding are poorly understood. Identifying these plant resistance factors would likely improve our understanding of plant–insect interactions and aid in developing highly resistant varieties.

Differences in the structure and relative abundance of vital intercellular organelles may also outline important traits conferring resistance. The physiological roles of chloroplasts and mitochondria in aphid resistance have been well-documented. Chloroplasts and mitochondria are responsible for eliciting Ca^{2+} signals in response to insect herbivory (Maffei et al. 2007). Organelle Ca^{2+} signaling occurs prior to gene expression and may define a host plant's ability to react quickly to an aphid aggressor (Maffei et al. 2007; Nomura and Shiina 2014). The size and orientation of these organelles may also indirectly affect the host plant's physiological capacity to withstand infestation (Jaipargas et al. 2015; Xiong et al. 2017). Large mitochondria have been correlated with high rate of protein transcription and translation (Miettinen and Björklund 2017). Moreover, alterations in mitochondrion form can be a direct reflection of the continuous flux in cellular energy (Jaipargas et al. 2015). A large mitochondrion may therefore have a greater capacity to perceive infestation and initiate induced plant defenses than a small mitochondrion. Moreover, chloroplast size has been theorized as an indication of photosynthetic efficiency, with large chloroplasts being less efficient compared to small chloroplasts (Jeon et al. 2002; Xiong et al. 2017). Therefore, a plant with small chloroplasts and a high photosynthetic capacity might withstand infestation more efficiently. These ultrastructural traits could be used as an important diagnostic tool in resistance screening.

Unknown resistance traits may provide objective determinants of resistance independent of visual observation and infestation, the latter being the conventional approach to resistance screening. These traits may also outline potential bottom-up explanations for antibiosis, antixenosis, and tolerance attributed to insect–plant interactions in sorghum. In this study, we address two objectives to elucidate specific mechanisms of resistance in sorghum according to leaf traits. Our objectives were: (1) to determine the key leaf structural traits mediating the physical innate resistance to SA in sorghum under no SA infestation; and (2) to identify indicative and non-indicative leaf structural traits of innate SA resistance. The leaf surface topography and mesophyll morphology of multiple species of aphid-resistant plants have been reported as resistance factors (Powel et al. 2006; Sharma et al. 2021; Leybourne and Aradottir 2022). We hypothesize: (1) epidermal and mesophyll leaf traits will be indicative of SA resistance in sorghum; (2) SA-resistant varieties will have a greater density of leaf trichome and stomata; and (3) epidermal traits will have a greater contribution to the pre-existing variation between resistant and susceptible varieties than mesophyll traits. Findings should provide a framework for a novel method of SA screening and an index of potential traits indicating resistance to aphid feeding in sorghum.

Materials and methods

Plant materials and growth conditions

We selected two SA-resistant (R. LBK1 and R. Tx2783) and two SA-susceptible (R. Tx7000 and R. Tx430) sorghum genotypes based on their different degree of SA tolerance. Singh et al. (2004), Armstrong et al. (2015), and Hayes et al. (2019) confirmed their source and degree of resistance. All samples were collected from uninfested conditions to ensure the identification of traits associated with innate SA resistance. We conducted controlled environment experiments (terminated at the five-leaf stage) to evaluate potential leaf structural and ultrastructural traits indicative of SA resistance at an early developmental stage. Traits were analyzed using light and transmission electron microscopy (TEM). For evaluating leaf structural surface traits at later stages (physiological maturity stage), leaf samples were collected from uninfested field experiments for scanning electron microscopy (SEM) analysis.

Controlled environment

Sixteen thermoformed nursery pots (4 replicates per genotype) were distributed in the growth chamber in a completely randomized design. Eight seeds per pot (16.5 cm in diameter and 17.0 cm in height) were sown in 1000 g of Miracle-Gro Enriched Potting Mix with Miracle-Gro Plant Food (Miracle-Gro Lawn Products, Inc., Port Washington, NY, USA). The seedlings were maintained under controlled environmental conditions (30/20 °C day/night temperatures with 14/10 light/dark hours). Seedlings were thinned at the three-leaf stage to four per pot and were maintained well-watered (80–90% soil field capacity) throughout the experiment.

Field experiment

The field experiment was conducted in Lubbock, TX (33.6° N, 101.9°W) at the Cropping Systems Research Laboratory of the USDA-ARS. Sorghum seeds (98,800 seeds/ha) were planted at 3 cm depth using a modified John Deere MaxEmerge Planter on June 5, 2021. This experiment comprised four single-row plots per genotype organized in a randomized complete block design (RCBD). Plots were 5.3 m in length and arranged with 1.0 m row spacings. Pre-plant irrigation plus subsurface drip irrigation of 4.22 mm/day was applied from planting to physiological maturity (Emendack et al. 2021). Weed control was manually conducted as needed in the plots.

Light and transmission electron microscopy (TEM)

We took the leaf samples from the growth chamber for TEM analysis at the five-leaf stage (the foremost fully expanded leaf). The rationale of utilizing a controlled environment was (1) to minimize the environmental effects and ensure changes were associated with only genotypic differences, and (2) to evaluate the feasibility of SA resistance screening at early plant growth stages using structural traits associated with innate SA resistance. Three biological replicates per genotype were collected at the five-true leaf stage. Two 2×1 mm leaf cuttings and one stem sample per biological replicate were excised with a razor blade under deionized water (DI) to maintain hydrated samples. All leaf cuts were parallel with the leaf's venation and equidistant between the midrib and leaf edge. Following collection, samples were immediately submerged in fixative [2% paraformaldehyde, 2% glutaraldehyde in 0.05 M cacodylate buffer (pH 7–7.2)]. The submerged leaf samples were refrigerated for 48 h before gently agitating for three h. The samples were then post-fixed in 2% osmium tetroxide for 2 h at 20 °C, dehydrated in a gradient ethanol series, and embedded in epoxy resin. For light microscopy, thick (1 µm) sections were cut using a glass edge on Power Tome-XL Ultramicrotome (USA), stained with 1% methylene blue and azure II, and examined using an Olympus BX41. For TEM, serial ultrathin (0.07–0.09 µm) sections were cut using a diamond knife on Power Tome-XL Ultramicrotome, stained with uranyl acetate and lead citrate, and examined with a Hitachi H-7650 transmission electron microscope. All images were analyzed using ImageJ software. Leaf thickness was determined by light microscopy. The leaf traits determined by TEM were epidermal cell wall thickness, cuticle thickness, intercellular width, mesophyll cell area, and chloroplast/mitochondria abundance and ultrastructure.

Scanning electron microscopy (SEM)

We took the SEM data from physiologically mature plants (leaf samples from the first fully expanded leaf under the flag leaf) from the field. SA never infested the test plot (validated by biweekly screening). The rationale for the target growth stage was based on results from previous studies in our lab indicating that the most significant variation in stomata and trichome traits occurs at physiological maturity (non-published data). Three biological replications (1×1 cm leaf cuttings) per genotype from different replicated plots were taken at physiological maturity (139 days after planting). Cuttings were collected from the first fully developed leaf beneath the flag leaf, excised with dissection scissors, and immediately submerged in 50 ml plastic vials with deionized water. The vials were placed on ice and transported to the lab for imaging. Sample collection and imaging were conducted following an adjusted SEM protocol developed in our lab (Laza et al. 2021). Samples were mounted on the sample stub-holder and imaged without pretreatment using the Hitachi S-4300 E/N (FESEM). All images were analyzed using ImageJ software. Leaf traits analyzed using SEM were trichome density, stomata size, and stomata density.

Statistical analysis

Analyses were conducted using genotype as a primary factor. All data were tested for homogeneity of variance and normality via PROC UNIVARIATE procedure in SAS 9.4 (SAS Institute, Cary, NC, USA). Data that were normally distributed were further analyzed using PROC GLM. Mean separation was executed with the least squares means procedure. Non-normally distributed data were analyzed using a non-parametric Kruskal–Wallis test and the Dwass, Steel, Critchlow–Fligner multiple comparison procedure when appropriate. A principal component analysis was performed on a subset of 10 traits indicative of resistance using JMP (JMP Statistical Discovery LLC, Cary, NC, USA). These traits were chosen by their ability to identify resistant genotypes from susceptible statistically. The data of all four sorghum genotypes were prepared as a correlation matrix for the ten traits. Identifying the traits responsible for the greatest variance orthogonal to the secondary variation enables a refined list of specific traits of interest.

Results

Leaf epidermal traits

The leaf surface topography between resistant and susceptible genotypes was distinctly different based on observations and statistically different among genotypes. Comprised both prickle and bi-cellular trichomes, the total trichome density (average number of trichomes per mm^2) was significantly different ($F_{3,8} = 18.47$, $P < 0.001$) across genotypes. Resistant varieties had 39% more total trichomes per mm^2 than susceptible ones (Fig. 1, Fig. 2). Prickle trichome density ($F_{3,8} = 4.80$, $P = 0.03$) and length ($F_{3,57} = 3.27$, $P = 0.03$) was also significantly different. Resistant varieties had an average of 80% more prickle trichomes per mm^2 (Fig. 1, Fig. 2). These prickle trichomes were an average of 17% longer than susceptible ones (Fig. 1, Fig. 3). Bi-cellular trichome density ($F_{3,8} = 4.24$, $P = 0.04$) and length ($F_{3,127} = 11.05$, $P < 0.0001$) were also

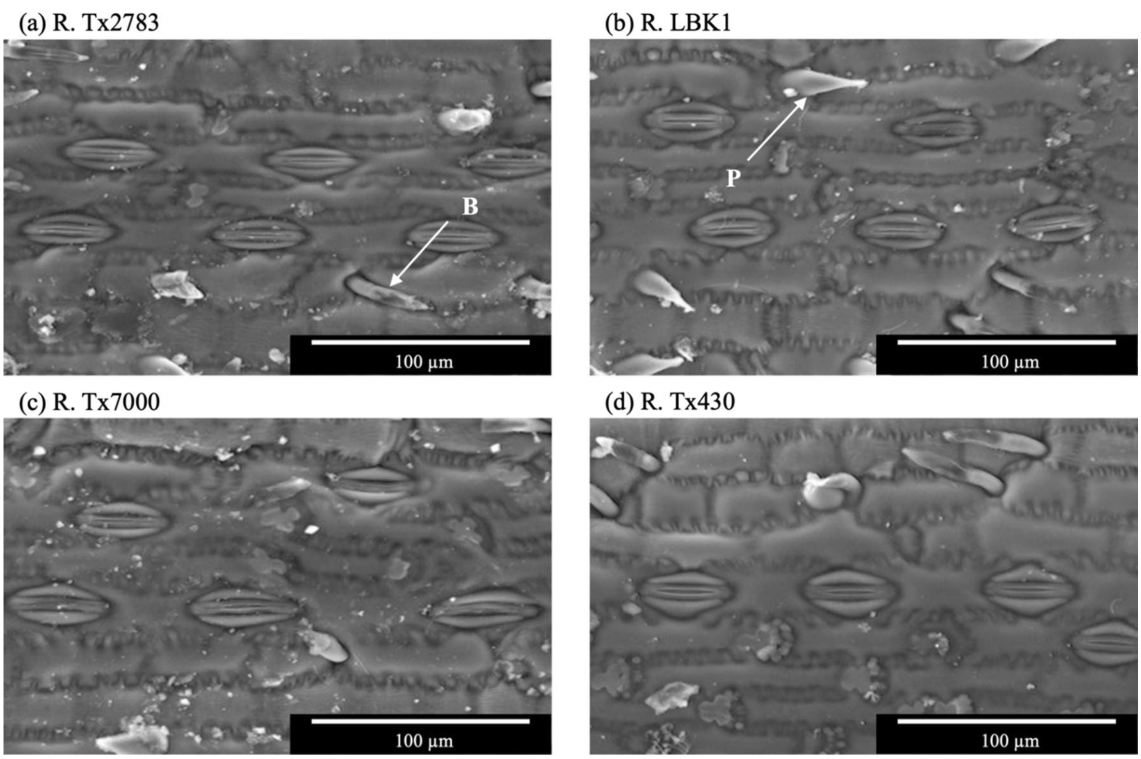

Fig. 1 SEM micrographs showing the stomatal and trichome densities on the abaxial leaf site among sorghum aphid resistant (**a**, **b**) and susceptible (**c**,**d**) genotypes. **a** R. Tx 2783 had the greatest stomata density. **b** R. LBK1 had the greatest trichome density. **c** R. Tx7000 had the lowest trichome density. **d** R. Tx430 had the lowest stomata density. Both, **a** and **b** genotypes, had the longest prickle (P) and bi-cellular (B) trichomes

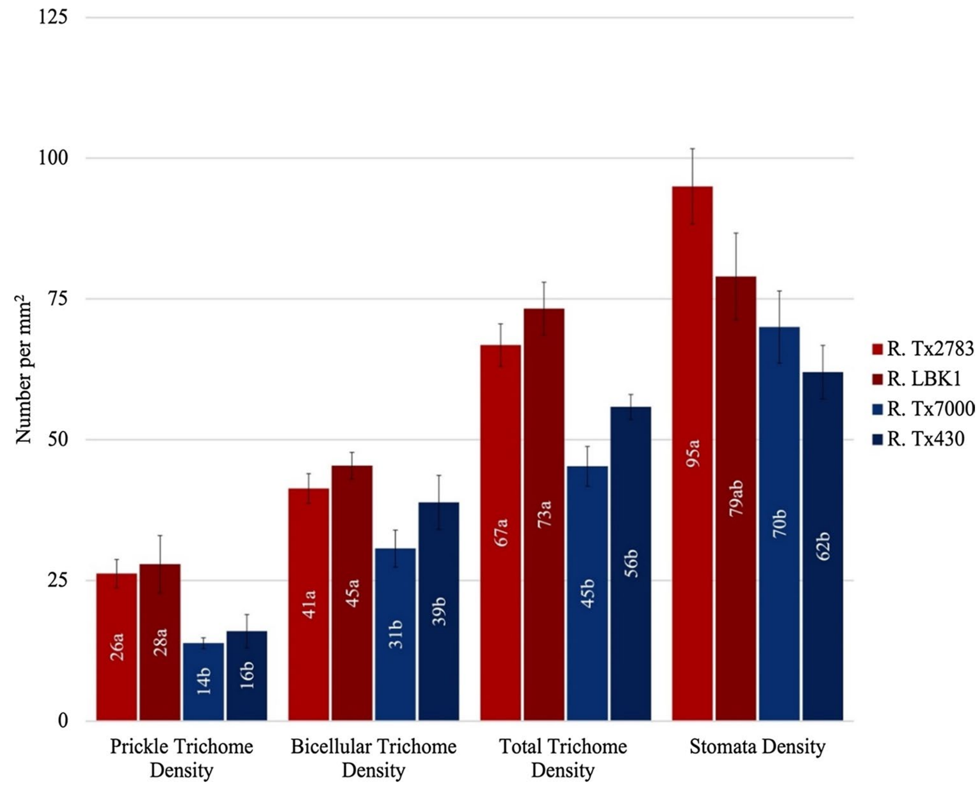

Fig. 2 Variation in the number of trichomes and stomata per mm^2 in sorghum aphid resistant (red) and susceptible (blue) genotypes. Resistant genotypes had 31% more stomata and 39% more total trichomes. Bars are means (± SE, $n=4$). *P* values are given for treatment comparisons with an ANOVA test followed by a least square means procedure. Lowercase letters (a-d) indicate significant difference ($P<0.05$) for assessed parameters

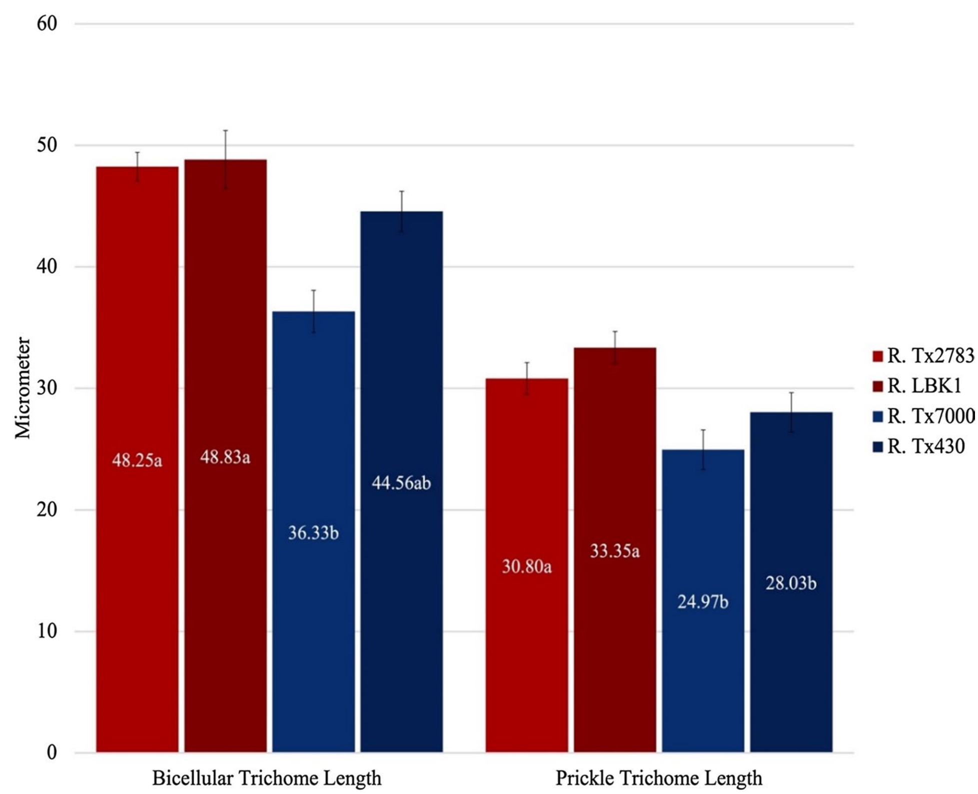

Fig. 3 Variation in trichomes length in sorghum aphid resistant (red) and susceptible (blue) genotypes. The prickle trichomes and bi-cellular trichomes of resistant varieties were 21% and 22% longer than susceptible, respectively. Bars are means (± SE, $n=4$). *P* values are given for treatment comparisons with an ANOVA test followed by a least square means procedure. Lowercase letters (a-d) indicate significant difference ($P<0.05$) for assessed parameters

significantly different across genotypes. Resistant varieties had 23% more bi-cellular trichomes per mm^2 than susceptible ones (Fig. 1, Fig. 2). These bi-cellular trichomes were an average of 22% longer (Fig. 1, Fig. 3). The ratio of prickle to bi-cellular trichomes in R. Tx2783 and R. LBK1 were 0.4 and 0.6, respectively. In contrast, susceptible R.

Tx430 and R. Tx7000 had a prickle to bi-cellular trichome ratio of 0.7 and 0.83, respectively (Table 1). Similar to total trichome density, resistant genotypes had significantly more stomata per mm^2 than susceptible genotypes ($F_{3,8}=4.71$, $P=0.03$). Resistant varieties had an average of 31% more stomata than susceptible ones (Fig. 1, Fig. 2). Despite the significance of stomatal density, there were no discernable differences in the length of the stomata ($F_{3,288}=0.38$, $P=0.77$; Table 1). Cuticle thickness also did not follow the same trend as stomata and trichome density ($F_{3,154}=0.11$, $P=0.11$). Epidermal cell wall thickness showed a significant pattern across genotypes (*chi-square* = 161.74, $P<0.0001$, $df=3$). The susceptible R. Tx430 exhibited the thickest epidermal cell wall thickness, followed by R. LBK1, R. Tx2783, and R. Tx7000 (Table 1, Fig. 4).

Table 1 Variation in the cuticle thickness, epidermal cell wall thickness, stomata, and trichomes in sorghum aphid resistant and susceptible sorghum genotypes

Parameter	Resistant		Susceptible	
	R. Tx2783	R. LBK1	R. Tx7000	R. Tx430
Leaf Thickness (µm)	98.34 ± 0.66d	101.17 ± 1.12c	110.23 ± 1.24a	105.82 ± 0.85b
Cuticle Thickness (nm)	68.45 ± 1.21	64.41 ± 0.77	68.60 ± 0.83	68.94 ± 1.10
Epidermal Cell Wall Thickness (µm)	0.89 ± 0.47b	1.24 ± 0.20c	0.82 ± 0.95a	1.57 ± 0.23d
Stomata size (um)	42.52 ± 0.45	42.49 ± 0.49	43.16 ± 0.76	42.37 ± 0.67
Bi-cellular:Prickle Trichome Ratio	0.40	0.60	0.83	0.70

Mean ± SE ($n=4$), lowercase letters (abcd): indicates significant difference ($P<0.05$) for assessed parameters

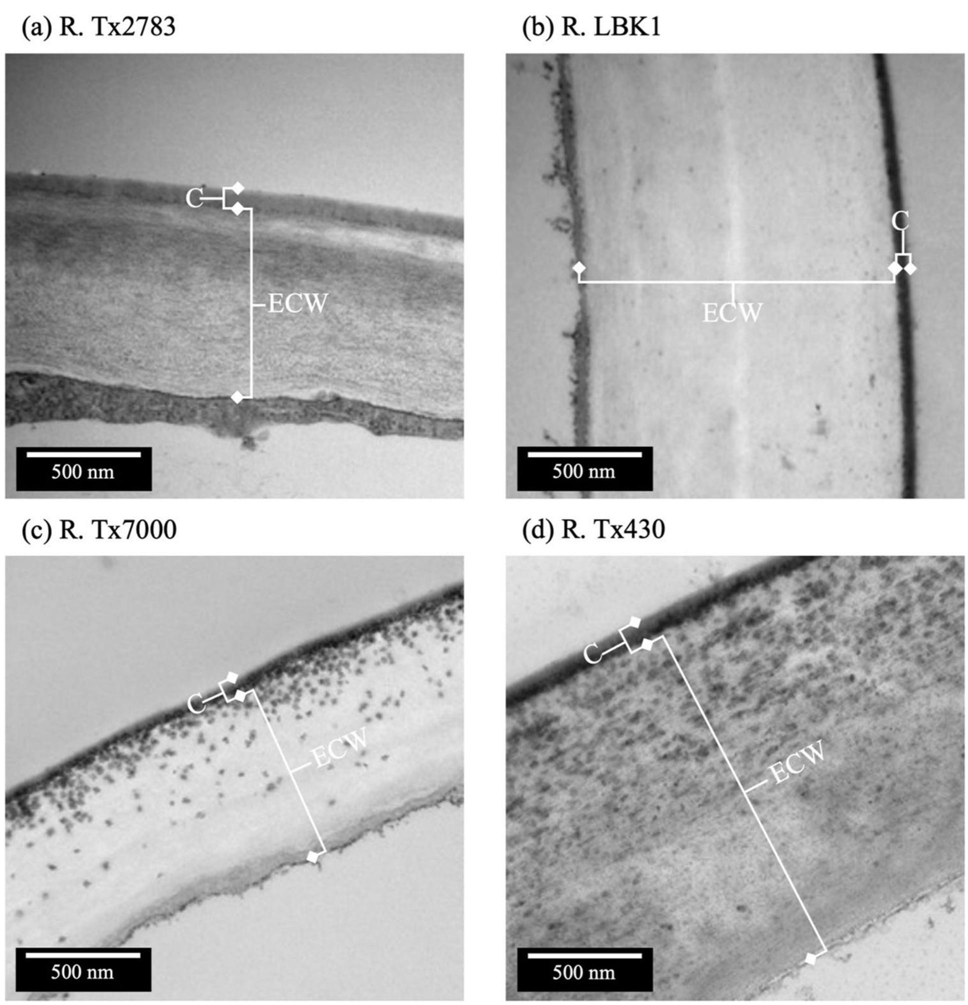

Fig. 4 TEM micrographs showing the variations in cuticle (C) and epidermal cell wall (ECW) thickness in sorghum aphid resistant (**a, b**) and susceptible (**c, d**) genotypes. R. Tx7000 had the thinnest epidermal cell wall followed by Tx2783, LBKk1, and Tx430

Leaf mesophyll traits

The width of these intercellular pathways notably differed between genotypes ($F_{3,196} = 1356.21$, $P < 0.0001$). The average apoplastic pathway of resistant varieties was 52% thinner than susceptible varieties (Figs. 5, 6). R. Tx2783 had an intercellular width that was 57% and 71% thinner than R. Tx430 and R. Tx7000, respectively (Figs. 5, 6). R. LBK1 followed a similar trend at 23% and 48% thinner than R. Tx430 and R. Tx7000, respectively (Figs. 5, 6). Leaf thickness was also significantly different ($F_{3,196} = 27.54$, $P < 0.0001$). The leaves of susceptible varieties were 9% thicker on average (Table 1). The susceptible R. Tx7000 exhibited the thickest leaves, followed by R. Tx430, R. LBK1, and R. Tx2783 (Table 1). Cell area significantly differed between genotypes ($F_{3,20} = 11.49$, $P < 0.0001$). The data suggest R. Tx7000 to have the largest mesophyll cells followed by R. LBK1, R. Tx430, and R. Tx2783 (Table 2, Fig. 7). R.Tx 7000 mesophyll cells were 1.34, 1.36, and 2.69 times larger than R. LBK1, R.Tx 430, and R. Tx2783, respectively (Table 2, Fig. 7). The chloroplast and mitochondria density, defined as the number of chloroplasts per 400 μm^2, were also distinctive traits ($F_{3,12} = 20.45$, $P < 0.0001$, $F_{3,12} = 4.81$, $P = 0.04$; Fig. 4). Resistant varieties had an average of 42% more chloroplasts and 18% more mitochondria per 400 μm^2 than susceptible (Table 2). The ratios of chloroplast to mitochondria in R. Tx2783 and R. LBK1 were 1.25 and 1.42, respectively (Table 2). The chloroplast-to-mitochondria ratios of R. Tx430 and R. Tx7000 however, were 1 and 1.16 (Table 2).

Leaf chloroplast traits

The chloroplast area and perimeter both followed distinct trends. The chloroplast area and perimeter of R. Tx7000 were significantly larger than all other genotypes ($F_{3,20} = 15.77$, $P < 0.001$, $F_{3,24} = 11.68$, $P < 0.001$; Fig. 8). The chloroplasts of R. Tx7000 had a 42% greater area and a 23% longer perimeter than all other varieties (Table 2). The length and width of R. Tx7000 chloroplast were also significantly different ($F_{3,20} = 4.32$, $P < 0.05$, $F_{3,24} = 5.05$, $P < 0.05$). R. Tx7000 chloroplasts were also 22% longer and 23% wider than all other varieties

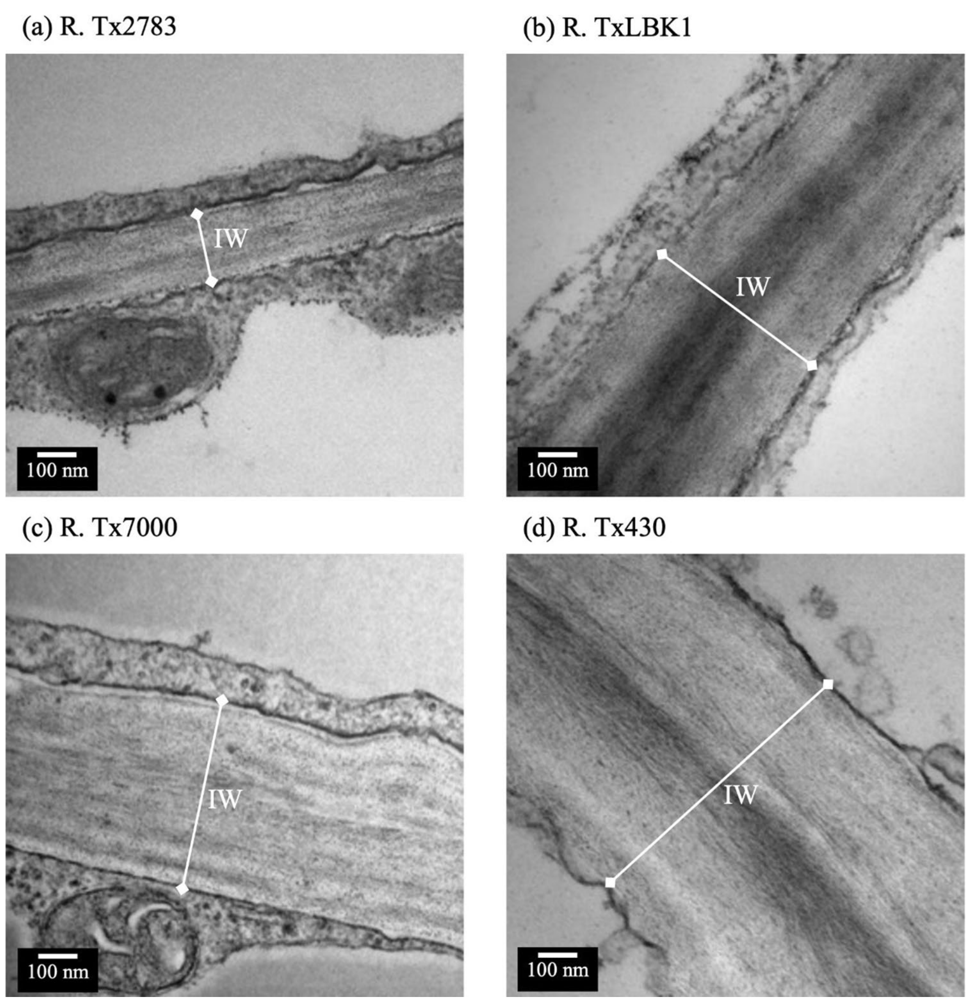

Fig. 5 TEM micrographs showing the differences in intercellular width (IW) in sorghum aphid resistant (**a, b**) and susceptible (**c, d**) genotypes. **a** R. Tx2783 had the thinnest IW followed by **b**, **c**, and **d**

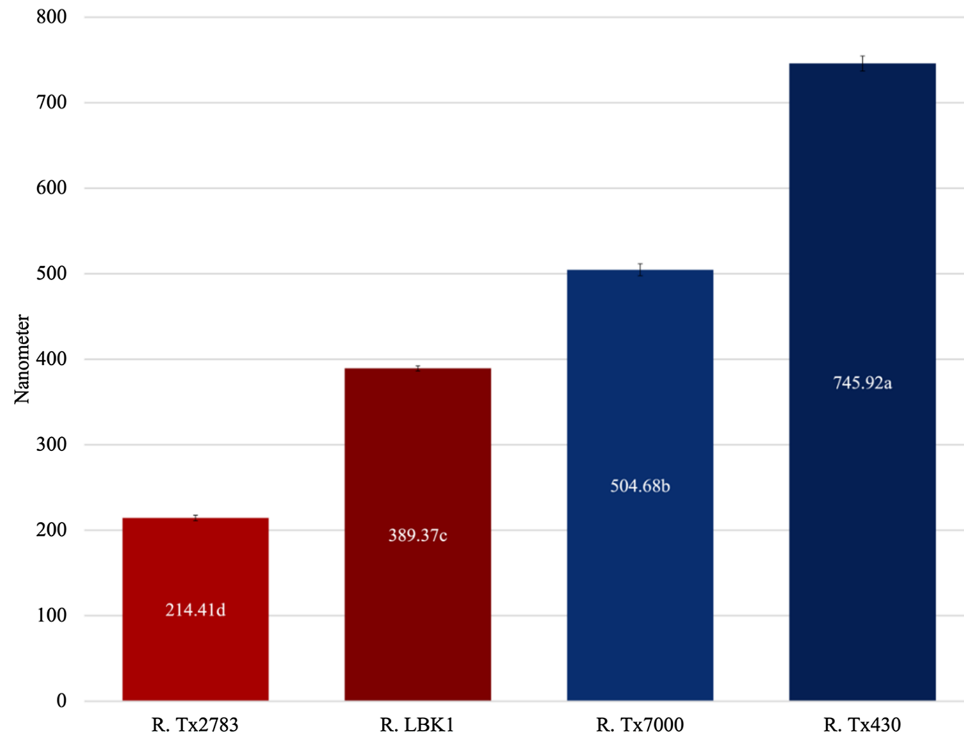

Fig. 6 Variations in mesophyll intercellular width in sorghum aphid resistant (red) and susceptible (blue) genotypes. Resistant genotypes were on average 52% thinner than susceptible. Bars are means (±SE, $n=4$). P values are given for treatment comparisons with an ANOVA test followed by a least square means procedure. Lowercase letters indicate significant difference ($P<0.05$) for assessed parameters. (1000 nm = 1 µm)

Table 2 Variation in mesophyll, chloroplast, and mitochondrion morphology in sorghum aphid resistant and susceptible sorghum lines

Category	Parameter	Resistant		Susceptible	
		R. Tx2783	R. LBK1	R. Tx7000	R. Tx430
Mesophyll morphology	Mesophyll cell area (µm^2)	87.70 ± 12.86c	179.50 ± 10.64b	237.29 ± 24.79a	191.59 ± 11.15b
	Chloroplast density (per 400 µm^2)	9.75 ± 1.10a	9.50 ± 0.96a	6.50 ± 0.28b	7.75 ± 0.49ab
	Mitochondria density (per 400 µm^2)	7.75 ± 0.48a	7.00 ± 0.95b	6.50 ± 0.87b	6.00 ± 0.41b
	Chloroplast: Mitochondria (per 400 µm^2)	10:81.25	10:71.42	7:71.00	7:6
Chloroplast morphology	Area (µm^2)	16.41 ± 1.33b	14.49 ± 1.50b	25.71 ± 1.48a	13.47 ± 1.29b
	Perimeter (µm)	15.99 ± 0.61b	14.78 ± 0.74b	19.67 ± 0.51a	14.92 ± 7.79b
	Length (µm)	4.87 ± 0.29ab	4.29 ± 0.26b	5.67 ± 0.28a	4.07 ± 0.49b
	Width (µm)	4.72 ± 0.35b	4.62 ± 0.36b	6.18 ± 0.24a	4.88 ± 0.33b
	Plastoglobuli (per chloroplast)	18.33 ± 1.48a	9.00 ± 1.00b	20.50 ± 1.93a	7.17 ± 0.79b
	Thylakoid (per 1 µm^2)	2.01 ± 0.08b	2.39 ± 0.12a	1.37 ± 0.08c	2.10 ± 0.22a
	Circularity	0.80	0.82	0.83	0.75
Mitochondria morphology	Area (µm^2)	0.27 ± 0.01c	0.28 ± 0.01b	0.30 ± 0.01b	0.34 ± 0.01a
	Perimeter (µm)	1.74 ± 0.05c	2.04 ± 0.05b	2.13 ± 0.05ab	2.25 ± 0.05a
	Length (µm)	0.50 ± 0.03c	0.62 ± 0.01ab	0.59 ± 0.02b	0.66 ± 0.02a
	Width (µm)	0.53 ± 0.02b	0.69 ± 0.03ab	0.68 ± 0.04a	0.61 ± 0.03a
	Circularity	0.85	0.86	0.83	0.85

Mean ± SE (n = 4), abc: indicates significant difference ($P<0.05$) for assessed parameters

(Table 2). Additionally, the number of plastoglobuli also differed between genotypes ($F_{3,20}=24.60$, $P<0.0001$). R. Tx7000 and R. Tx2783 had a combined average of 55% more plastoglobuli per chloroplast than R. LBK1 and R. Tx430 (Table 2, Fig. 8). The number of grana significantly differed ($F_{3,20}=8.86$, $P<0.001$). R. Tx7000 had an average of 36% fewer thylakoids per 1 µm^2 than all other genotypes (Table 2). The shape of the chloroplast also exhibited a distinct trend. However, it failed to meet significance standards ($F_{3,20}=3.074$, $P=0.0511$). Coinciding with chloroplast, the structure of the mitochondria was also different among genotypes.

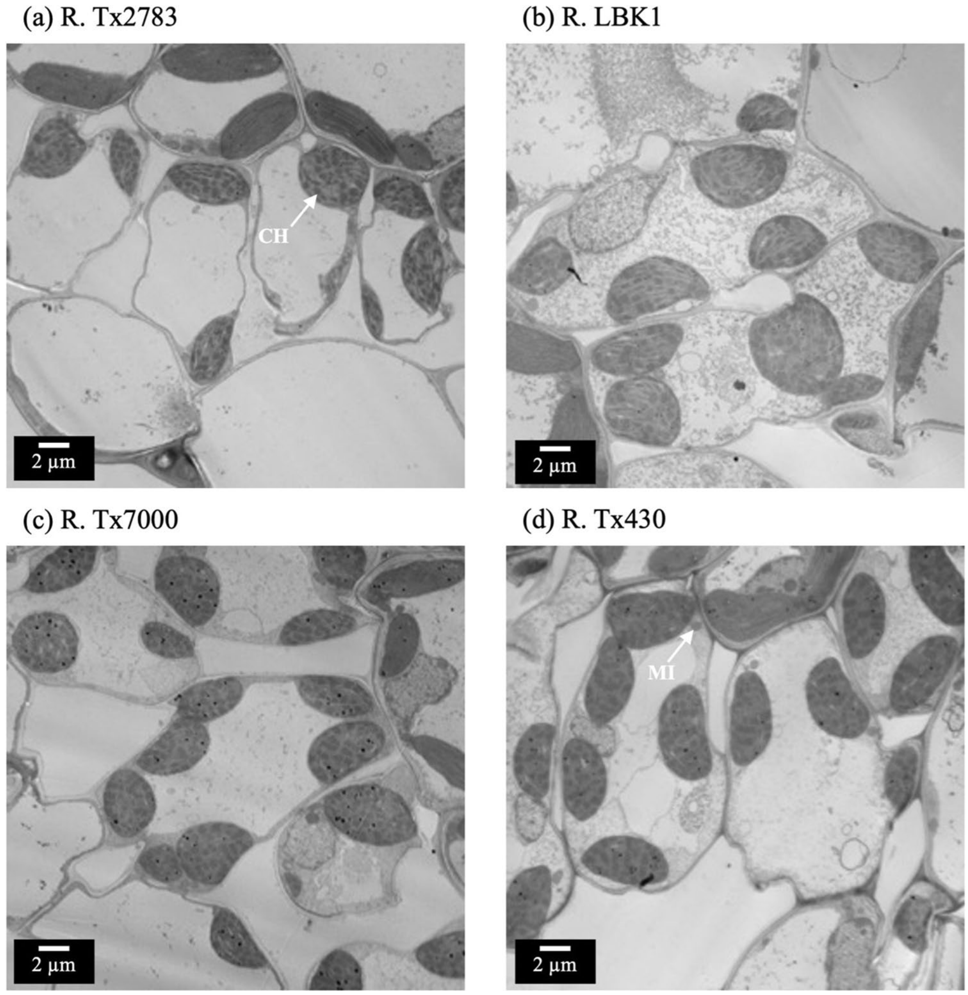

Fig. 7 TEM micrographs showing the variations in mesophyll cell morphology, chloroplast density (CH), and mitochondria density (MI) in sorghum aphid resistant (**a**,**b**) and susceptible (**c**, **d**) genotypes. **a** R. Tx2783 had the smallest cell area followed by genotypes **b**, **d**, and **c**. Resistant genotypes **a** and **b** had an average of three more chloroplasts per 400 μm^2. **a** R. Tx2783 had one more mitochondrion per 400 μm^2 than all other genotypes

Leaf mitochondrial traits

The area occupied by mitochondria was a noteworthy trait ($F_{3,20}=18.05$, $P<0.0001$). The average area of mitochondria from resistant varieties was 14% smaller than susceptible (Table 2). Mitochondria perimeter followed a similar trend ($F_{3,20}=17.21$, $P<0.0001$). Resistant varieties expressed a 14% shorter mitochondrial perimeter (Table 2). Mitochondrial length and width were also significantly different ($F_{3,20}=9.25$, $P<0.001$, $F_{3,20}=6.37$, $P<0.001$). The mitochondria of the resistant R. Tx2783 were 20% shorter and 18% narrower than susceptible varieties (Table 2, Fig. 9). The resistant R. LBK1 mitochondria, however, were 0.8% shorter and 7% wider than susceptible ones (Table 2, Fig. 9). The circularity of the mitochondria was similar across genotypes ($F_{3,20}=0.65$, $P=0.59$).

Principal component analysis and biplot investigation of traits indicative of resistance

The first four components contributed 97.35% of the total variation. Principal component one (PC1) alone contributed 69.77% of the total variation. Trichome density, bi-cellular trichome length, prickle trichome length, and stomatal density contributed to the most variation with 0.37, 0.35, 0.32, and 0.31, respectively (Table 3). Prickle trichome density exhibited the lowest contribution (0.23) in PC1 (Table 3). Leaf thickness, intercellular width, and mesophyll cell counterbalanced the observed variation and did not contribute to the observed variation (Table 3). Principal component two (PC2) accounted for 16.90% of the total variation. Prickle trichome density, bi-cellular trichome density, and intercellular width were primarily responsible for the observed variation and contributed 0.45, 0.43, and 0.48, respectively (Table 3). PC1 and PC2 contributed over 75% of the variation; therefore, these are the only two axes interpreted. The biplot graph indicated the variability of the four genotypes for the ten traits under study. Each genotype is separated into different quadrants (Fig. 10). The genotype's position relative to PC2 was not indicative of SA resistance. Their position relative to PC1, however, could separate resistant from susceptible. Resistant genotypes were oriented in quadrants one and four, indicating a positive direction along PC1

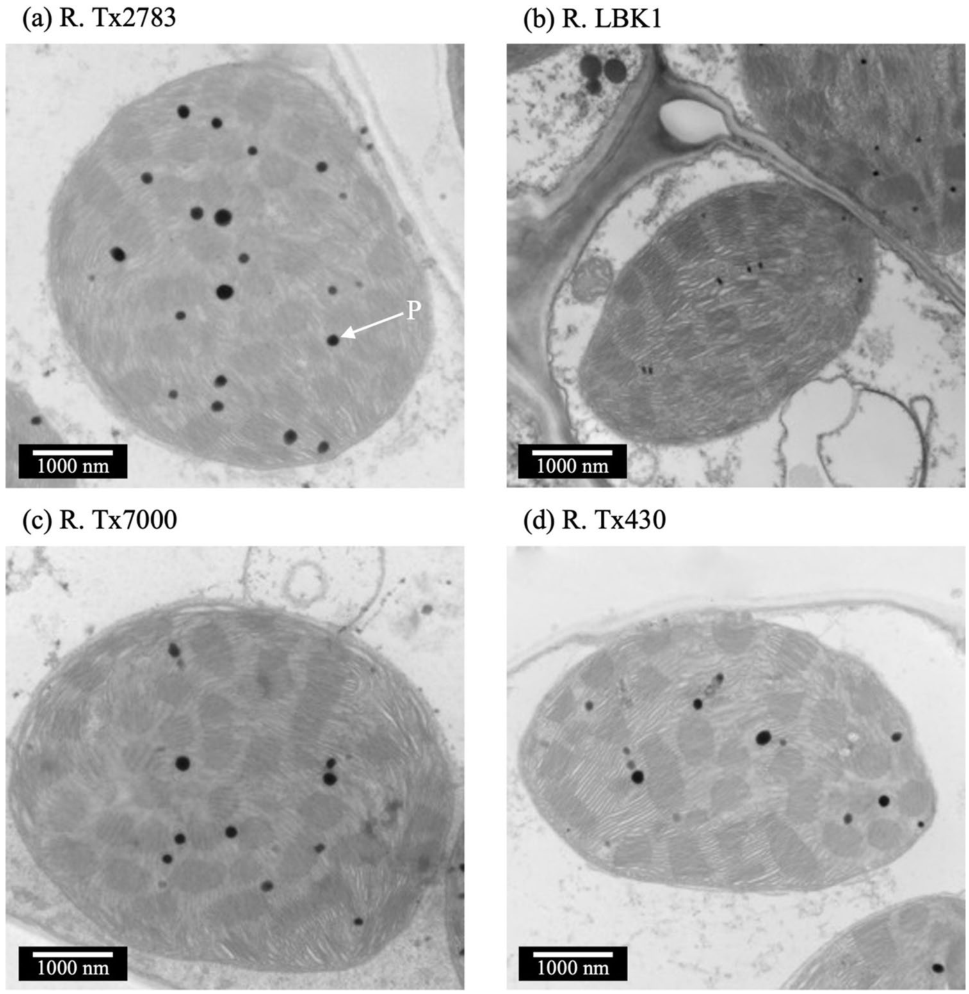

Fig. 8 TEM micrographs showing the chloroplast morphology in sorghum aphid resistant (**a, b**) and susceptible (**c, d**) genotypes. **c** R. Tx7000 had the largest chloroplast. R.Tx2783 (**a**) and R. Tx7000 (**c**) had the most plastoglobuli (P) per chloroplast

(Fig. 10). Susceptible genotypes were in quadrants two and three, indicating a negative direction along PC1 (Fig. 10). Trait vectors may also be observed following distinctive trends. Excluding stomata density, the vectors of all traits were positively oriented (Fig. 10). Intercellular width, mesophyll cell area, and leaf thickness were the only traits that indicated a negative direction (Fig. 10). Furthermore, all trichome-related traits were within the first quadrant, indicating a positive orientation along the PC1 and PC2.

Discussion

Our findings suggest significant quantifiable differences between resistant and susceptible Sorghum varieties. Ten out of the 28 observed traits may be linked to SA resistance. These traits include stomata density, trichome densities, trichome lengths, leaf thickness, mesophyll cell area, chloroplast density, and intercellular width. The 21 remaining traits failed to separate resistant from susceptible materials. These traits focused on the ultrastructure of the chloroplast, mitochondria, stomata, epidermal cell wall, and cuticle. Nevertheless, our findings display ten innate traits that distinguish uninfested resistant from susceptible varieties.

Resistant varieties had longer trichomes and greater trichome density

Resistant varieties had an average of 43% more total trichomes than susceptible ones. Trichome density could be directly augmenting the settling behavior of the aphid upon individuals settling on the leaf. Aphids are less likely to settle in areas with trichomes. This behavior has been shown in multiple studies (Levin 1973; Webster et al. 1994; Peter et al. 1995). Certain cultivars of cotton, alfalfa, potato, wheat, and tomato have been dubbed aphid resistant in direct correlation with trichome density (Goffreda et al. 1989; Peter et al. 1995). However, the definitive role of trichomes in aphid resistance has been a subject of intense debate.

In contrast, zero ascertainable physical barriers were found in sorghum in correlation to corn leaf aphid, *Rhopalosiphum maidis*, resistance (Kariyat et al. 2019). The

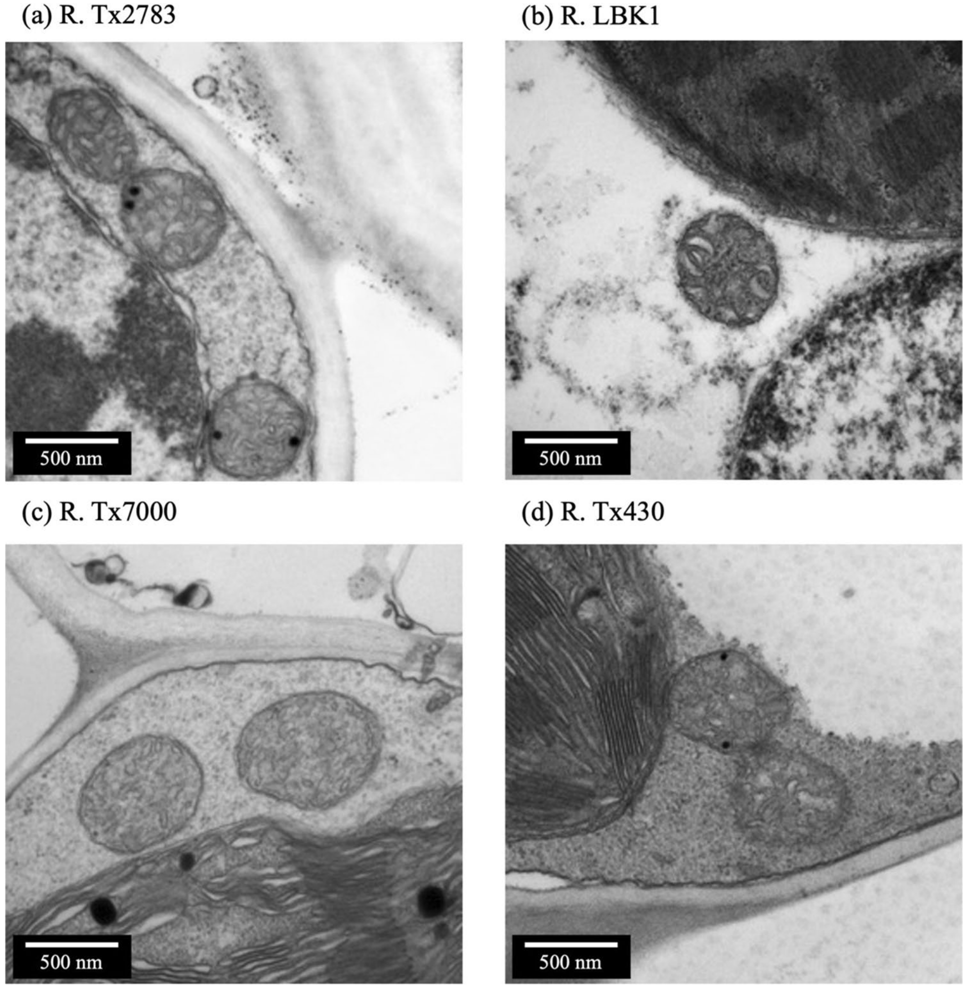

Fig. 9 TEM micrographs showing the mitochondrial morphology in sorghum aphid resistant (**a, b**) and susceptible (**c, d**) genotypes. **c** R. Tx7000 had the largest mitochondria of all other genotypes. **a** R. Tx2783 had comparatively smaller mitochondria than all other genotypes

Table 3 Contribution percentages associated with the first four principle components and the respective Eigen vectors

Parameter	PC1	PC2	PC3	PC4
Eigen value	6.98	1.69	0.74	0.33
Difference	5.29	0.95	0.41	0.19
Proportion	69.77	16.90	7.37	3.31
Cumulative	69.77	86.67	94.04	97.35
Eigen vectors				
Prickle trichome length	0.31	0.22	− 0.12	0.82
Bi-cellular trichome length	0.35	0.04	− 0.344	0.04
Stomata density	0.31	− 0.40	0.134	− 0.17
Trichome density	0.36	0.21	− 0.11	− 0.22
Prickle trichome density	0.23	0.45	0.59	− 0.16
Bi-cellular trichome density	0.30	0.43	− 0.10	− 0.37
Leaf thickness	− 0.37	0.15	0.12	0.12
Intercellular width	− 0.24	0.48	− 0.53	− 0.09
Mesophyll cell area	− 0.32	0.34	0.37	0.14
Chloroplast density	0.35	0.02	0.23	0.21

authors did not identify trichome density as a resistance factor. To improve microscopic resolution, they coated the leaf samples with golf prior to imaging. This practice likely concealed the trichomes present on the surface of the leaf. Nevertheless, similar analyses conducted on soybeans found trichomes to be a neutral factor (Faiz et al. 2021; Nalam et al. 2021). Considering the narrow host range of the soybean aphid, *Aphis glycines*, these results support the notion of an ecological niche shared by specialist herbivories (Blackman and Eastop 2000; Chen et al. 2017). Specialist herbivores have been shown to overcome, augment, and even exploit plant resistance factors to their advantage (Zhang et al. 2019). This response may explain the neutral effect of trichomes against soybean aphids. Our data suggest total trichomes density as a potential trait indicative of SA resistance in sorghum. The phenotyping field trials affirm these conclusions, as the resistant materials were the last plots to be infested. R. LBK1 and R. Tx2783 were not infested until susceptible materials had been eliminated.

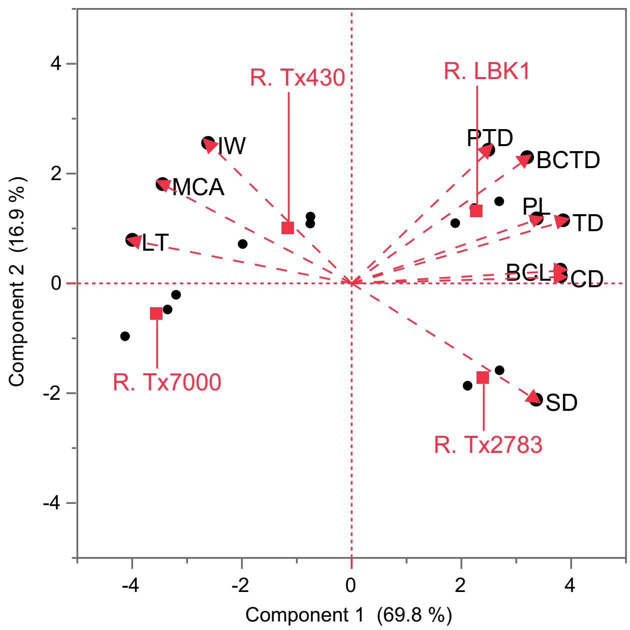

Fig. 10 Genotype by trait biplot graph of the first two principal components. The trait codes are: PL, Prickle Trichome Length; BCL, Bi-Cellular Trichome Length; SD, Stomatal Density; TD, Trichome Density; PTD, Prickle Trichome Density; BCTD, Bi-Cellular Trichome Density; LT, Leaf Thickness; IW, Intercellular Width; MCA, Mesophyll Cell Area; CD, Chloroplast Density

The resistant R. Tx2783 and R. LBK1 possessed longer bi-cellular and prickle trichomes. Although the bi-cellular trichome density was an indistinguishable trait among genotypes, prickle trichome density was shown to be a good SA resistance indicative trait. Furthermore, the ratio of bi-cellular trichomes to prickle trichomes disclosed that both R. Tx2783 and R. LBK1 had a higher proportion of prickle trichomes on the leaf's surface. R. LBK1 had the greatest prickle trichomes densities across genotypes. These findings suggest that the influence of trichomes on aphid resistance is more dynamic than previously reported.

Prickle trichomes are non-glandular protrusions along the cell margins. Aphid behavior may be influenced by trichomes. For example, prickle trichomes may entangle and or impale arthropods attempting to move along the surface (Khan et al. 2000; Sarria et al. 2010; Riddick and Simmons 2014; Karabourniotis et al. 2020). Moreover, a greater degree of nymphal mortality, reduced aphid longevity, and fecundity has been correlated with the presence of non-glandular trichomes (Xing et al. 2017). R. LBK1 had the longest prickle trichomes and the greatest density. Prickle trichomes found on R. Tx2783 were similar in size, yet fewer in number. These results further affirm our hypothesis that a major component of R. LBK1 resistance is within the boundaries of antixenosis. We also hypothesize that prickle trichome's influence on SA resistance is both density- and size-dependent. The greater the prickle trichome density, the more likely aphid may become entangled or impaled. However, the length of the prickle trichome could also affect the extent of entanglement and impalement. These assumptions partially explain host preference and the aphid's archetypal decline in fitness and fecundity on sorghum aphid-resistant plants. Further research is necessary to quantify the degree of influence of prickle trichomes on sorghum aphid behavior, fitness, and fecundity.

Bi-cellular trichomes, like prickle trichomes, are located along the cell margin. They are classified as glandular trichomes in sorghum. Capable of discharging a mucilage-type material, the exudate of bi-cellular trichomes found within the genus *Sorghum* has yet to be fully elucidated (McWhorter et al. 1993, 1995). Histological staining of the secretory product of the bi-cellular trichomes of *S. halepense* suggest a high concentration of carbohydrates, predominantly callose (McWhorter et al. 1995). Callose is commonly associated with a rapid response to injury and sieve element occlusion. Phloem-located defense mechanisms and aphid-feeding deterrents have been frequently correlated with callose (Will and van Bel 2006; Kim et al. 2020; Twayana et al. 2022). Although the study on *S. halepense* failed to conduct a histological stain on *S. bicolor* trichomes, an electron micrograph did clearly indicate the bi-cellular trichomes of *S. bicolor* to closely resemble those found on *S. halepense* (McWhorter et al. 1995). The micrograph also indicated *S. bicolor* bi-cellular trichomes to exude a similar mucilage-type material (McWhorter et al. 1995). Further research is needed to fully understand the chemical composition of bi-cellular trichomes on sorghum and their influence on aphid behavior and fitness.

Resistant varieties had greater stomatal density

The two resistant genotypes had an average of 31% greater stomatal density than the susceptible. R. Tx2783 had the greatest stomatal density, followed by R. LBK1, R. Tx7000, and R. Tx430. In agreement with our results, stomatal density was a good indicator of tolerance to the English grain aphid, *Sitobion avenae,* and bird cherry oat aphid, *Rhopalosiphum padi,* on wheat (Luo et al. 2019). A greater stomatal density will also alter the leaf surface topography and elicit an insect's behavioral response (Ibbotson and Kennedy 1959; Powell et al. 1999, 2006). Considering these assumptions, our findings further support the assertion of resistant sorghum germplasm maintaining comparable levels of photosynthesis while infested (Paudyal et al. 2020).

Resistant varieties had narrow mesophyll intercellular width

Our results showed that the intercellular width of the resistant varieties was 52% thinner than that of susceptible varieties. Previous work reported intercellular space as a component of resistance (Aravind and Kajjidoni 2007; Saska et al. 2021). Mesophyll resistance factors have also been postulated to be a significant component of aphid resistance (Schwarzkopf et al. 2013; Leybourne et al. 2019; González et al. 2022). This broad sweeping category has been linked to multiple abnormal behaviors identified by electrical penetration graphs. Excessive probing, salivating, and time until feeding have all been attributed to mesophyll resistance factors. R. Tx2783, and presumably R. LBK1, has been shown to prevent continuous feeding and elicit probing behavior at a greater degree (Tetreault et al. 2019). The effects of these feeding behavior alterations may be observed as the customary decrease in fecundity and longevity associated with resistant genotypes. Therefore, the apoplast's width may partially explain both antibiosis and tolerance. A population of sorghum aphids on resistant germplasm will increase at a slower rate (Limaje et al. 2018; Paudyal et al. 2019). This slow increase will allow the plant a greater opportunity to compensate for infestation. The apoplast distance is not the only factor influencing this reaction. Nevertheless, we hypothesize that the narrow intercellular distance of resistant germplasm influences aphid feeding efficacy negatively.

Resistant varieties had smaller mesophyll cells and greater chloroplast density

Resistant varieties had smaller mesophyll cells and 43% greater chloroplast density.

These findings further support a difference in mesophyll architecture and cell size between resistant and susceptible genotypes. Smaller cells have been theorized to promote cell cycling, photosynthetic efficiency, and rigidity (De Micco et al. 2011; Holland et al. 2020). The utility of small cells in relation to aphid resistance is likely correlated with mesophyll resistance factors. Mesophyll, with many small cells, will exhibit a network of intricate intercellular pathways arranged in a tight jigsaw pattern. In contrast, large-celled mesophyll exhibits straightforward wide intercellular paths toward the vascular bundle. This correlation between cell size and aphid resistance has been observed in wheat. A smaller number of large mesophyll cells, displaying simple intercellular pathways, was theorized to influence the susceptibility of a wheat landrace to the bird cherry-oat aphid, *Rhopalosiphum padi* (Singh et al. 2020). This mesophyll trait was not a reliable indicator of susceptibility across all varieties. Certain susceptible varieties exhibited the same mesophyll phenotype as the resistant ones (Singh et al. 2020). The influence of mesophyll architecture and thereby chloroplast density, on aphid resistance has yet to be fully elucidated. Our study found that these resistant genotypes consistently exhibited smaller mesophyll cells and more chloroplasts.

Leaf structural traits non-indicative of innate SA resistance in sorghum

In our study, the chloroplast, mitochondria, and epidermal cell ultrastructural traits were ineffective indicators of SA resistance. However, there were a few noteworthy findings. Mesophyll cell size, intercellular width, and organelle size in sorghum were interconnected. The smaller the cell, the narrower the intercellular width and the smaller the chloroplast and mitochondria. R. Tx2783 had characteristically small mesophyll cells, very narrow intercellular spaces, and small organelles, while R. Tx7000 had the largest mesophyll cells, wide intercellular spaces, and the largest organelles. These differences may be linked to gas exchange, water use efficiency, and even cell signaling efficacy.

In this study, plastoglobuli distribution and epidermal cell wall morphology did not differ between resistant and susceptible genotypes. The two genotypes with the most plastoglobuli per chloroplast were the susceptible R. Tx7000 and resistant R. Tx2783. These finding suggest plastoglobuli may be correlated with plant and grain color. Both R. Tx7000 and R. Tx2783 have a red-plant necrotic lesion color and red grain. R. Tx 430 and R.LBK1 have a tan-plant necrotic lesion color with yellow and white grain, respectively. In this context, such traits have been associated with stress in other plant species (van Wijk and Kessler 2017). Although these traits cannot be directly used for screening SA-resistant genotypes, they offer a unique perspective into the conservative microanatomy within the species. The structural basis of these vital organelles has yet to be fully understood.

Stomatal density, total trichome density, and trichome length contribute the most variation

The first four principal components together exerted 97.35% of total variation through stomata density, trichome density, trichome lengths, leaf thickness, mesophyll cell area, intercellular width, and chloroplast density. Mesophyll cell area, intercellular width, and chloroplast density are not easily quantified traits. Although these may be important components of resistance, there is an evident bottleneck that will prevent efficient screening. Those traits indicative of the leaf's surface, may be implemented into high-throughput screening programs. The first principal component, responsible for 69.77% of the total variation, validated this proposition. Stomatal density, total trichome density, and trichome lengths contributed the most variation, whereas the traits indicative of the mesophyll, excluding chloroplast density, had no significant contribution.

The PCA effectively separates the selected genotypes based on their degree of resistance. R. Tx7000 and R. Tx430 are plotted in quadrants two and three, respectively. Both quadrants indicate a negative direction along PC1. R.

Tx7000, the most susceptible variety, is plotted at the furthest negative point from zero. The second most susceptible variety, R. Tx430, is plotted closer to the center of the plot. The same trend can be observed for R. LBK1, and R. Tx2783. These resistant varieties are plotted in quadrants that indicate a positive direction along PC1. According to their position along PC1, R. LBK1 and R. Tx2783 are similar, indicating comparable levels of resistance. Our assumption is that the distance from zero, in either direction along PC1, will ultimately disclose the inherent resistance of the variety when considering these factors. The field phenotyping conducted by our team in 2021 and 2022 confirm the degree of resistance of the selected varieties (not published data), and agree with previous studies (Sharma et al. 2013; Armstrong et al. 2015).

Conclusion

Significant structural differences exist between the SA-resistant and the susceptible sorghum genotypes evaluated. We identified innate characteristics under non-SA-infested conditions. The key traits responsible for the most critical degree of genotypic variation and indicative of innate physical resistance include stomatal density, trichome density, length, and chloroplast density. We suggest screening a diverse sorghum germplasm panel to validate our findings. Further research on the interplay between morpho-anatomical, structural, physiological, and biochemical traits will provide new insights into the SA resistance mechanism in sorghum. Our findings provide the foundation for developing an objective SA resistance screening method that requires no SA infestation.

Author contribution statement ET and HL conceived and designed the study. ET drafted the document with HL assistance. ET took measurements and acquired data. ET prepared and imaged leaf samples. ET analyzed the data with assistance of YE and HL. ET, HL and YE assisted with the field experimental design, field operations. All authors contributed to: (1) analysis and interpretation of data, (2) revising the manuscript critically; and (3) approved the final version to be submitted.

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Data availability All data used and analyzed during this study are included in this article.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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